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Title: Berry-Esseen bounds for design-based causal inference with possibly diverging treatment levels and varying group sizes.

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This paper provides a long-needed theoretical unification of design-based causal inference under Neyman’s randomization model, accommodating settings where the number of treatment levels Q diverges, and group sizes N_q vary widely—including cases with only one observation per arm. Prior theory either fixed Q or required replications in every arm, leaving modern experiments such as high-dimensional factorial designs, conjoint surveys, and partially nested provider studies without rigorous justification. The authors derive Berry-Esseen bounds (BEBs) for linear estimators and quadratic forms, propose novel variance estimators for unreplicated designs, and establish conditions for valid Wald-type inference.

The core model is classical. For N units and Q treatment levels, complete randomization assigns exactly N_q units to level q . Unit i has fixed potential outcome $Y_i(q)$. The target parameter is $\gamma = F^\top \bar{Y}$, where $\bar{Y} = (\bar{Y}(1), \dots, \bar{Y}(Q))^\top$ are population means and F is a prespecified $Q \times H$ contrast matrix. The natural estimator is $\hat{\gamma} = F^\top \hat{Y}$ with $\hat{Y}_q$ the sample mean in arm q . The authors study the standardized statistic $\tilde{\gamma} = V_{\hat{\gamma}}^{-1/2}(\hat{\gamma} - \gamma)$ and the quadratic form $T = (\hat{\gamma} - \gamma)^\top V_{\hat{\gamma}}^{-1}(\hat{\gamma} - \gamma)$.

The paper’s logical development proceeds through three key innovations. First, the authors formulate $\hat{\gamma}$ as a linear permutation statistic and apply Stein’s method via exchangeable pairs. Theorem 1 gives a general BEB for linear projections of $\tilde{\gamma}$ with two complementary bounds: one effective when all N_q are large, and another that trades off between arms with many units (term I) and arms with many small groups (term II). This second bound is essential when Q

diverges, but individual N_q are bounded.

Second, the authors specialize these bounds to nearly uniform designs (all N_q within constants of a common N_0) and general designs (partitioning arms into large Q_L , small replicated Q_R with $2 \leq N_q \leq \bar{n}$, and unreplicated Q_U with $N_q = 1$). Under mild conditions on F —specifically $\|F\|_\infty = O(Q^{-1})$ and $\rho_{\min}(F^\top F) \geq cQ^{-1}$ —Corollaries 1 and 2 show that the BEB for linear projections shrinks at rate $\sqrt{H/N}$ for nearly uniform designs, and as the maximum of term I over large arms and $\sqrt{H/N_S}$ over small arms for general designs, where N_S is the total sample size in small arms. These rates reveal a fundamental trade-off: local replication reduces error for individual arms, while averaging over many arms reduces error globally.

Third, the paper addresses the severe challenge of variance estimation when $N_q = 1$ and the usual sample variance is undefined. The authors propose two strategies. The first is a global estimator that is conservative but often strictly so. The second and more refined strategy groups unreplicated arms into clusters of size at least two (Definition 4), constructing $\hat{V}_{\hat{Y}}(q, q) = \mu_{\langle g \rangle}(Y_q - \hat{Y}_{\langle g \rangle})^2$ for each arm q in group $\langle g \rangle$. Lemma 2 and Theorem 7 demonstrate that this estimator is conservative under mild conditions on within-group correlations and mean heterogeneity, with estimation error in operator norm bounded by $O_{\mathbb{P}}(\|F\|_\infty^4 QH^4)$ under bounded fourth moments.

The significance of this work is considerable. It provides the first non-asymptotic, uniform BEBs that explicitly track dependence on Q , H , and the heterogeneity of N_q , thereby rigorously justifying Wald inference in 2^K factorial designs with K growing with N . The variance estimators for unreplicated designs open valid inference in previously intractable settings such as experiments with exactly one observation per treatment level. Theorems 4 and 11 show that Wald-type confidence sets are asymptotically valid provided $H^{19/2}/N \rightarrow 0$, an extremely mild condition covering most practical applications. Overall, this paper delivers a foundational advance for design-based causal inference, equipping both theorists and practitioners with rigorous finite-sample guarantees for modern, complex experimental designs.